QUALITY ASSURANCE · TEST PLAN

VWO Login Dashboard

Authentication, recovery, accessibility, security, and performance

Product: VWO at app.vwo.com **Plan version:** 1.0 **Prepared:** 22 October 2026

Source: Supplied Product Requirements Document

> **Planning note:** QA and Pre-Prod URLs, named project owners, calendar dates, and formal acceptance thresholds were not supplied. They remain TBD and require confirmation. The two-sprint duration follows the supplied test-plan template.

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Objective

Verify that the VWO login experience provides secure, reliable, accessible, and efficient entry to the VWO platform, while supporting the authentication and account-recovery journeys defined in the PRD.

Testing will assess the login form and its validation, session handling, remember-me behavior, optional multi-factor authentication (2FA), enterprise single sign-on (SSO), password recovery, responsive presentation, accessibility, integrations, and stated performance and security requirements.

Scope

In Scope

- Email and password authentication, field validation, loading feedback, and actionable errors.
- Remember-me behavior, secure session creation and timeout, and successful transition to the VWO dashboard.
- Password reset and recovery, including secure reset-token behavior and password requirements.
- Optional 2FA; enterprise SSO protocols (SAML, OAuth, and other supported methods); and optional social identity providers, when enabled for the release.
- Responsive layouts, light/dark themes, keyboard and screen-reader access, clickable labels, and WCAG 2.1 AA requirements.
- HTTPS and authentication-data protection, rate limiting, compliance/audit expectations, analytics, support, and onboarding integrations.
- Login-page load target, high availability, concurrent login capacity, and geographic performance requirements.

Out of Scope / Conditional

- Biometric authentication, adaptive authentication, and Progressive Web App enhancements are listed as future enhancements and are outside this plan unless added to the release scope.
- Social login and individual SSO providers are conditional on the providers and configurations selected for the release.
- Testing of the wider experimentation, personalization, and analytics product is limited to the login transition and integrations identified in the PRD.

The PRD does not specify detailed API contracts, supported password rules, session timeout values, supported identity providers, load profiles, or test data. These require product/engineering clarification before detailed test-case sign-off.

Inclusions

Authentication and session management

- Valid and invalid email/password combinations; empty fields; malformed email; field feedback on blur; password requirements and strength feedback.
- Successful login, failed login messaging, loading state, success confirmation, and transition to the personalized dashboard.
- Remember-me selection and persistence; configurable session timeout; secure token generation, handling, and expiration.
- Optional 2FA enrollment/challenge and recovery behavior; enterprise SSO flows; and configured social login flows.
- Rate-limiting behavior and protection against repeated failed authentication attempts.

Password recovery and user journeys

- Forgot-password request, email-based recovery, reset-token validity/expiry, password update, and return-to-login behavior.
- New-user registration/free-trial link and returning-user access, including preservation of prior-session context where supported.
- Error identification, recovery paths, support assistance, and clear completion confirmation.

Experience, accessibility, and compatibility

- Responsive and touch-friendly controls; autofocus; clickable labels; VWO branding; light/dark modes; and loading feedback.
- Keyboard-only operation, focus visibility/order, screen-reader labels/announcements, high-contrast presentation, and WCAG 2.1 AA evaluation.
- Browser, operating-system, and device combinations listed under Test Environments.

Security, integrations, and non-functional requirements

- HTTPS enforcement, encrypted authentication-data transmission, secure password storage practices, session security, access control, and sensitive-data exposure checks.
- GDPR/CCPA data-handling expectations, enterprise audit requirements, and OWASP authentication guidance identified in the PRD.
- VWO platform handoff; login success/failure analytics; customer-support integration; and configured identity-provider and onboarding/marketing integrations.
- Page-load target under 2 seconds on standard connections, 99.9% uptime, support for thousands of simultaneous login attempts, and multi-region behavior.
- Regression, smoke, sanity, retest, exploratory, integration, usability, security, performance, load, and accessibility testing.

Test Environments

Environment	URL / status	Purpose
QA	TBD by project team	Primary functional, integration, accessibility, security, and regression testing.
Pre-Prod	TBD by project team	Release-candidate verification and approved performance/security validation.
PRD reference	https://app.vwo.com/login	URL identified in the PRD; use of this live endpoint for testing must be explicitly authorized.

Platform	Browser	Coverage note
Windows 10	Chrome, Firefox, Microsoft Edge	Desktop compatibility and keyboard/accessibility coverage.
macOS	Safari	Desktop compatibility and keyboard/accessibility coverage.
Android	Chrome	Responsive layout and touch interaction.
iPhone / iOS	Safari	Responsive layout and touch interaction.

Specific OS/browser versions, device models, screen sizes, network profiles, hardware requirements, test accounts, and authentication credentials are TBD. Use approved non-production accounts and environment-specific identity-provider configuration.

Defect Reporting Procedure

1. Log each reproducible deviation from the approved PRD or signed-off test case in JIRA.
2. Include a concise title, environment and build, browser/device, preconditions, exact reproduction steps, expected and actual results, severity/priority, and relevant screenshots, logs, or request identifiers. Do not attach passwords, reset tokens, or other secrets.
3. Triage with the appropriate product, frontend, backend, or DevOps owner; assign severity and priority; and record status, resolution, and retest results in JIRA.
4. Share a daily end-of-day defect/status summary with development management, following the process in the supplied template.
5. Retest fixes and run related regression coverage before closure.

Tool: JIRA. **Project assignees:** TBD. The supplied template lists Devesh (New Frontend), Sonal (Backend), and Prajeeth (Dev Ops); confirm whether these contacts apply to this project before assigning defects.

Test Strategy

Planning and design

Create prioritized scenarios and test cases from the approved PRD, clarifying unspecified requirements before sign-off. Apply equivalence partitioning, boundary-value analysis, decision tables, state transitions, use-case testing, error guessing, and exploratory testing.

Execution approach

1. Perform smoke testing on each candidate build. If critical login paths fail, report the blockers and hold in-depth execution until a stable build is available.
2. On a stable build, execute prioritized functional, recovery, integration, compatibility, accessibility, security, and non-functional tests against the approved environments.
3. Run supported environments in parallel where multiple test resources are available; record environment-specific results.
4. Report defects in JIRA, retest fixes, and repeat regression/test cycles until agreed release criteria are met.

Testing principles

Context-driven **Shift-left** **Exploratory** **End-to-end** **Risk-prioritized**

Metrics and guardrails

Measure from PRD	Target / guardrail	Use in testing
Login success rate	95% or higher	Track successful authentication attempts against the agreed measurement definition.
Login page load	Under 2 seconds on standard connections	Measure page-load performance under the agreed standard connection profile.
Availability	99.9% uptime	Review availability evidence and monitoring for the agreed measurement window.
Concurrent login attempts	Thousands simultaneously	Load-test using a profile and exact volume to be agreed with engineering.
User satisfaction	90% or higher	Review user feedback using the product team's defined survey/measurement method.
Login-related support tickets	20% reduction	Compare against an agreed baseline and measurement window.
Security guardrails	Zero successful brute-force attacks or unauthorized access; zero session-hijacking incidents; 100% security-audit compliance	Record security test and audit outcomes; escalate any guardrail breach.

These are PRD success metrics, not automatically approved test exit criteria. Measurement definitions, baselines, load profiles, and release gates require stakeholder agreement.

Test Schedule

Planned duration: two sprints, as stated in the supplied template. Sprint dates and team capacity are TBD.

Task	Timing	Dependency / output
Test plan and requirement clarification	Sprint 1; dates TBD	Approved plan; confirmed environments, requirements, and acceptance thresholds.
Test scenario and case creation	Sprint 1; dates TBD	Prioritized scenarios and test cases for review/sign-off.
Smoke and test execution	Sprints 1-2; dates TBD	Build availability and approved QA/Pre-Prod access.
Retesting, regression, and summary report	Sprint 2; dates TBD	Fixes available; test and defect reports ready for closure.

Test Deliverables

- Approved test plan and clarified requirement/assumption log.
- Prioritized test scenarios, test cases, and traceability to PRD requirements.
- Execution results, environment/build details, and supporting evidence.
- JIRA defect records, retest results, and daily defect/status summaries.
- Security, accessibility, compatibility, integration, and performance/load findings relevant to the release.
- Final test summary report, open-risk/defect status, and approval record.

Entry and Exit Criteria

Requirement Analysis

Entry: Testing team receives the PRD or project requirements.

Exit: Requirements have been explored and understood by the testing team; questions and ambiguities are resolved or recorded with owners.

Test Design

Entry: Requirements are available for review.

Exit: Scenarios and test cases are prepared, prioritized, reviewed, and signed off by the client/stakeholders as required by the supplied template.

Test Execution

Entry criteria

- Test scenarios and cases are signed off by the client/stakeholders.
- Application build and approved test environment are available.
- Required accounts, test data, identity-provider configuration, and access permissions are ready.
- Smoke testing passes for the build before in-depth execution.

Exit criteria

- Planned test execution is completed or remaining coverage is documented with rationale.
- Test case and defect reports are ready; fixes selected for retest have been retested.
- Open defects, risks, metric results, and any unmet PRD guardrails are communicated for release decision.

> The PRD does not define pass-rate, defect-severity, or residual-risk thresholds for release. Stakeholders must approve these before execution exit/sign-off; no thresholds are inferred here.

Test Closure

Entry criteria: Test case and defect reports are ready.

Exit criteria: Test summary report is completed, open items and risks are documented, deliverables are submitted, and required stakeholder approvals are recorded.

Tools

- JIRA for defect tracking.
- Mind map tool for scenario exploration.
- Screenshot capture tool for evidence.
- Word and Excel documents for plans, cases, and reports.

Performance, security, and accessibility tooling is not named in the supplied PRD/template; selection and approval are TBD.

Risks and Mitigations

Risk	Mitigation / response from supplied material
Security weakness, unauthorized access, or non-compliance	Regular security audits and penetration testing; real-time security monitoring/alerts; regular security patches and vulnerability assessments. Escalate any breached security guardrail.
Performance degradation or traffic spikes	Load testing under agreed conditions; real-time performance monitoring/alerts; auto-scaling infrastructure as identified in the PRD.
Test environment or build URL unavailable	Use the supplied template response: resources work on other tasks while access/build availability is restored.
Insufficient testing time	Use the supplied template response: dynamically ramp resources based on client needs.
Required test resource unavailable	Backup resource planning, as stated in the supplied template.
Unspecified requirements, environment URLs, identity-provider setup, or metric definitions	Resolve or record as a dependency before test-case sign-off and relevant execution; project owners are TBD.

Approvals

Submit the following for client/stakeholder approval: test plan, test scenarios, test cases, and reports. Testing proceeds to the next applicable phase after required approval is received, following the supplied template.

Approval item	Approver	Status / date
Test Plan	TBD	Pending
Test Scenarios	TBD	Pending
Test Cases	TBD	Pending
Test Summary / Reports	TBD	Pending